

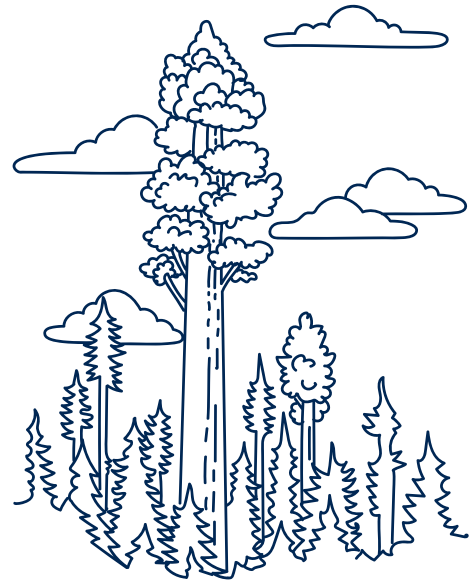
UVic Libraries - Community of Practice on AI application

Introduction to Prompt Engineering for GenAI



Lunchtime Session - June 6, 2024 - 12pm



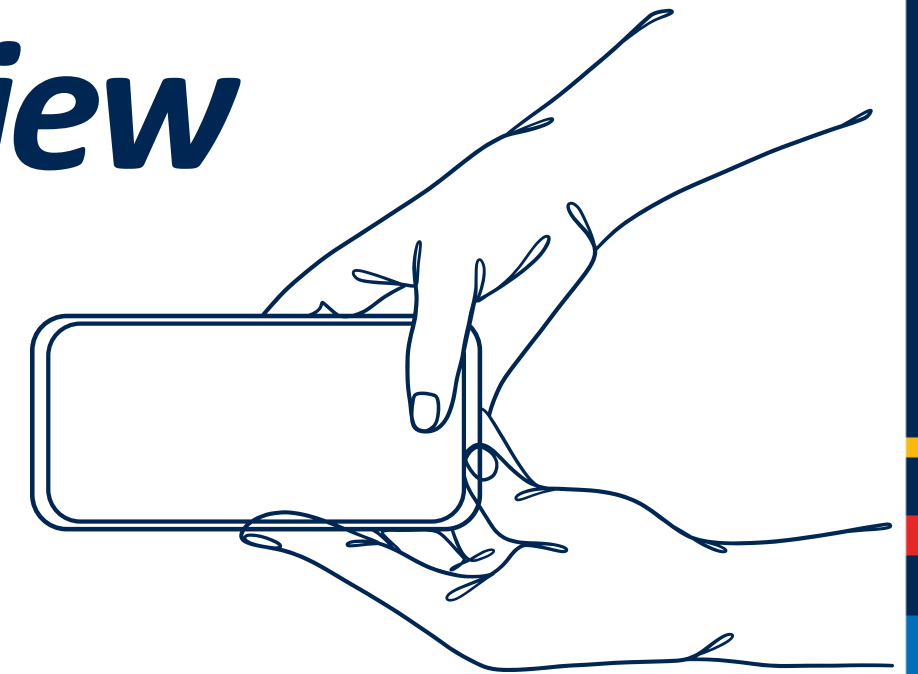


Territorial Acknowledgment

We acknowledge and respect the Ləkʷəŋən
(Songhees and Esquimalt) Peoples on whose territory
the university stands, and the Ləkʷəŋən and W̱SÁNEĆ
Peoples whose historical relationships with the land
continue to this day.

Prompt Engineering - Overview

12–12.45pm



Questions & Discussion

12.45–1pm



Objectives

After this session, you should be able to...

- Know basic prompt design principles
- Be familiar with prompting techniques
- Use prompts to address GenAI pitfalls
- Align prompting skills with academic integrity
- Have a sense on how to use this in your library work

Agenda

- 1) Role Of Prompts In GenAI
- 2) Principles Of Prompt Design
- 3) Prompt Engineering Techniques
- 4) Using Prompts To Address GenAI Pitfalls
- 5) Prompts & Academic Integrity
- 6) Questions?

Abbreviations & Terminology

- GenAI = *Generative AI*
- AI, GenAI, GenAI tool, AI model = *used synonymously here*
- Prompt Chains = *‘Conversations’ with GenAI; Prompt sequences*
- Prompt Design vs. Prompt Engineering
 - *terminological ambiguity*
 - *using them synonymously*
 - *‘Prompt Engineering’ most established*

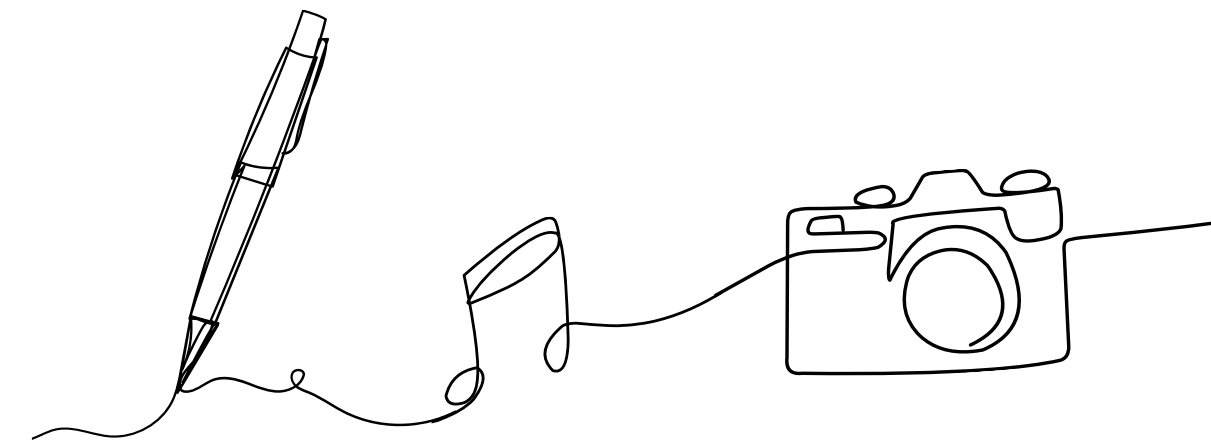


Role of Prompts in GenAI

Refresher: What is Generative AI?

Emphasizing aspects relevant for Prompt Engineering

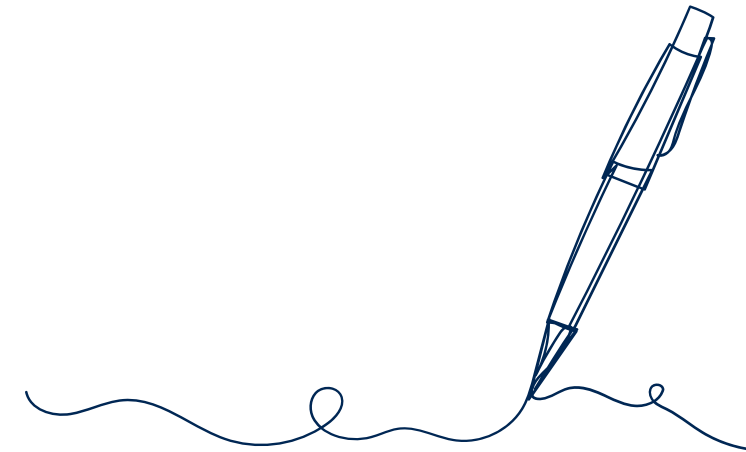
- New type of AI application
- Generates novel content on demand, using training data
- Unprecedented, adapting capabilities
- Capable of semantic & conceptual understanding
- Learns and evolves further
- Now most tools: multimodal
- Market segment with hyper-evolution

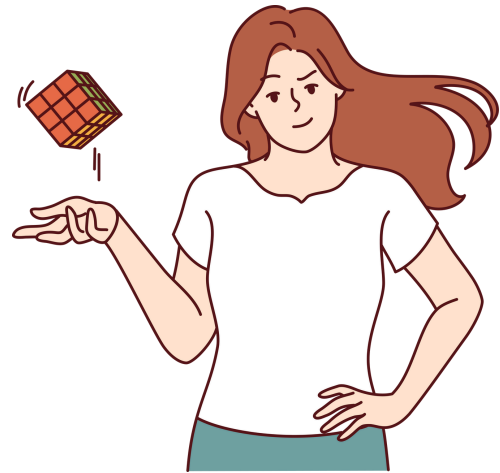


What is a Prompt?

- Input to an AI model in natural language
 - AI model understands query semantically
 - Then performs desired task

→ *keyword searches are NOT prompts*
- Via text interface, API, ...or microphone
- Sequence of prompts: *Prompt Chains!*





Benefits & Limitations



- Describe tasks in NL
→ *precisely & creatively*
- No expert skills needed
→ *e.g. Boolean; Code*
- Multi-lingual inputs

- Limited no. words per prompt
- Limited prompts per chain
- Lack of transparency
- Lack of repeatability

...many more

What is Prompt Engineering?

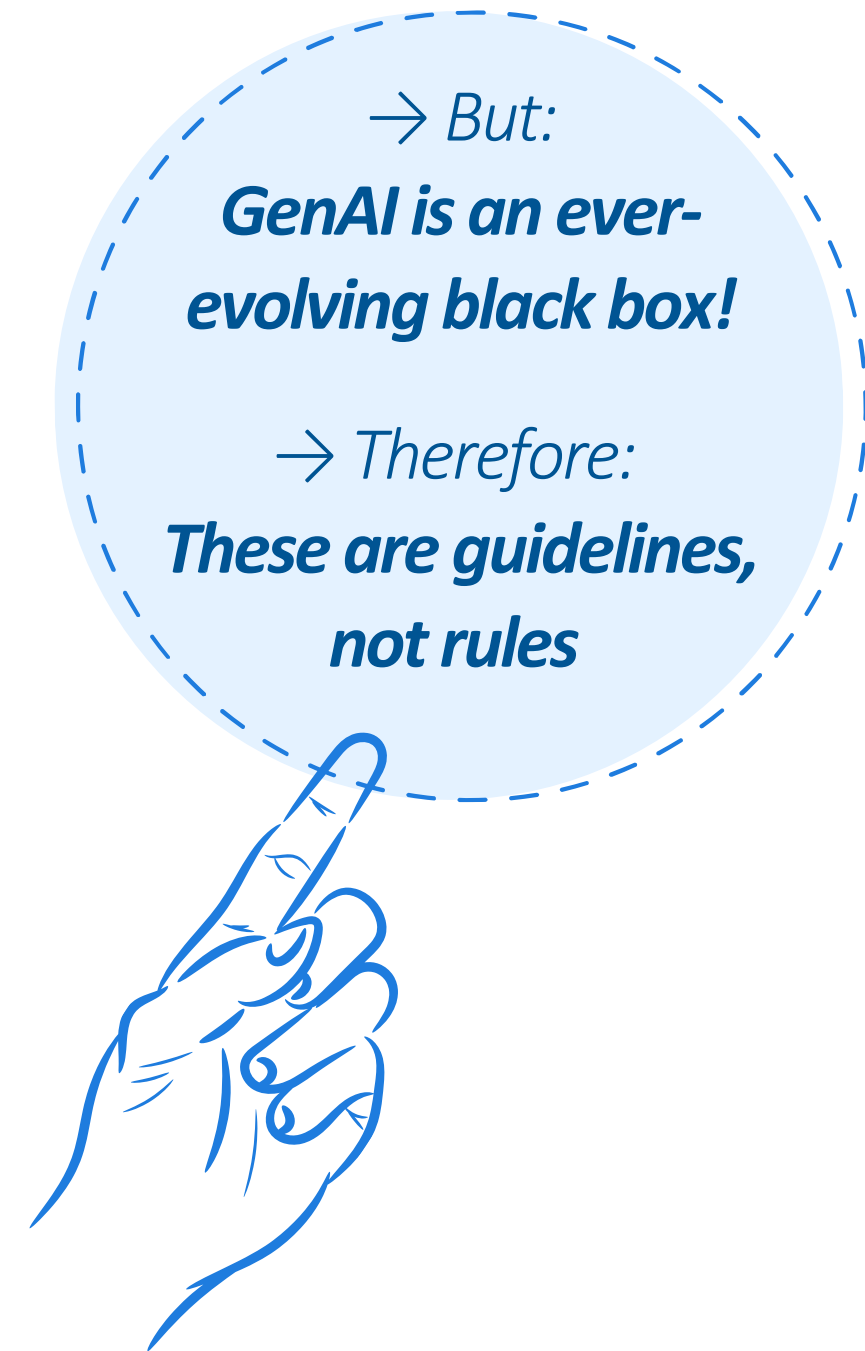
Role of Prompts in GenAI

= Craft of guiding GenAI to produce desired outcomes, efficiently using natural language instructions.

- Foundational AI literacy skill
- Synonyms: Prompting; Prompt Design

5 Purposes:

- 1: Maximize consistency; improve reliability & repeatability
- 2: Leverage an AI model's sophisticated capabilities
- 3: Individualize GenAI outcomes → from generic to specific
- 4: Address GenAI-related pitfalls
- 5: Safety-tests; Jailbreaking





7 Principles For Good Prompt Design

0

Most important one?

Be creative & have fun!

1 Be concise.

- Aim for brevity & clarity
- Use language that is as simple as possible

- **Cluttered prompt & question format**

“Can you provide me with a detailed explanation of the photosynthetic process and the significance of this biochemical reaction?”

EXAMPLE

- **More concise prompt & instructive format**

“Explain the process of photosynthesis and its significance in detail.”

2 Be clear.

- Avoid vague & ambiguous wording
- Be as specific as necessary

- **Prompt in ambiguous wording and question format:**

“How do I produce a paper?”

- **Clearer, less ambiguous and imperative format:**

“Explain the common steps of writing a research paper for a peer reviewed academic journal.”

EXAMPLE

3 Include context & logical structure.

- Provide context to improve reasoning
- Build structured & coherent prompts

- **Unstructured prompt without context:**

“Mention tasks involved in writing a research paper.”

EXAMPLE

- **Structured prompt, including context:**

*“List the steps involved in writing a research paper for a student assignment.
Begin with selecting a topic, end with proofreading the final draft.”*

4 Break down complex tasks.

In one prompt, avoid...

- combining several tasks
- asking for various formats

Divide multi-step or -topic tasks...

- into separate prompts
- or prompt chains

EXAMPLE

*"**Design** a comprehensive marketing campaign for a new eco-friendly product, detailing the target audience, key messaging, and promotional strategies. **Provide specific examples** of advertising mediums such as social media, print, and television, and create a visual mockup of an advertisement in a modern minimalist style. **Write** meaningful alt-text for the advertisement, summarizing its content for enhanced accessibility. Finally, **draft** a 2-minute pitch script for presenting this marketing campaign to potential investors."*

5 Specify desired output.

- Style & tone
- Depth & length
- Format, language, or content type
- Temperature → controls precision & creativity

- **Style & Tone:**

“Draft a paragraph on XYZ in the simplistic style of a tabloid paper. Use a sensational tone.”

- **Depth & length:**

“Provide a high-level legal argumentation on the topic of XYZ that does not exceed 500 words.”

- **Format, language, content type:**

For example: A three-column table; a poem in Swahili; a script in Python

- **Temperature:**

Balance creativity & precision, depending on task

EXAMPLE

6 Reflect & adapt your approach.

- Continuously evaluate outcomes
- Be flexible, fine-tune, & improve
- Adjust, then re-run!

- By applying the other principles
- By using a different prompting technique

EXAMPLE

→ *No improvement? Even with adjustments? --- **Start fresh!***

7 Combine these principles.

- These principles complement each other
- Mix and match modularly for optimum results

A red rectangular stamp with a distressed, ink-like texture. The word "EXAMPLE" is written in bold, uppercase, sans-serif letters, tilted slightly upwards to the right.

- **Earlier example, already improved (tone and style):**

“Draft a paragraph on climate change in the simplistic style of a tabloid paper. Use a sensational tone.”

- **Improved further: clarity, context, temperature, length:**

“Draft a newspaper paragraph of 500 words including a headline, on the effects of climate change. Apply the simplified grammar of a tabloid paper, and use sensational language. Be creative when describing climate change’s effects.”



5 Techniques for Efficient Prompts



1 Provide examples.

- Useful examples support reasoning
- Can relate to input or output
- Explain them, if necessary

EXAMPLE

- **Prompt without added example:**

“Make a claim on nutrition that might be seen as controversial by health experts.”

- **Same prompt, improved by an example:**

“Make a claim on nutrition that might be seen as controversial by health experts, for example ‘Chocolate is good for you’.”

2 'Role play': Use personas & scenarios.

- You are the movie director --- the AI is your method actor
- “Program” the AI model through pretend roles & scenarios
- Shortcut to fine-tuning tone, depth, style, ***even training data applied***

*“I am a **university student**, preparing for a very busy exam season. I have access to a GenAI chatbot. **You will be my instructor**, who is open-minded on teaching methods and new technology, but firm on academic integrity. **You are aware of** a recent spike in student breach of academic integrity standards through unauthorized GenAI tool use. **What advice would you give** my class on the usage of those tools, **that will convince me** not to use them, despite the time and effort I could save, and my conviction that nobody would catch me?”*

EXAMPLE

2 Generate or improve prompts with AI

- **Task the AI to write a prompt for you**
 - *Meta-prompting: Describe desired outcome, application, purpose, etc.*
- **Refine your own prompts using AI**
 - *Run your own prompt through AI → Ask for feedback and improvements*
 - *Be mindful and reflect*
- **Ask the GenAI tool, if & how it understands your prompt**
 - *Ask for an annotated example output*
 - *Or instruct the tool to explain which steps it would carry out, based on your prompt*

→ **CAUTION!**

*Some tools improve
human prompts
automatically*

→ *Be aware of the
implications!*



EXAMPLE

→ examples in the hands-on-activities

4 Mark-up your input

- **Simple punctuation already helps**
 - *but using 'fake' mark-up syntax is even more effective*
- **Standardize inputs**
 - *especially useful for tools that 'remember'*
- **Create re-usable templates**

Example → next page

...mark-up your input:

A red rectangular stamp with a distressed, ink-like texture. The word "EXAMPLE" is written in bold, uppercase, sans-serif letters inside the stamp.

- Earlier prompt, improved through specifying output parameters:

“Draft a newspaper paragraph of 500 words including a headline, on the effects of climate change. Apply the simplified grammar of a tabloid paper, and use sensational language. Be creative when describing climate change’s effects.”

- Improving it further, by marking-up facets of it

*“**Task:** Draft a newspaper paragraph of 500 words including a headline. **Topic:** Effects of climate change. **Style:** Use the simplified grammar of a tabloid paper. **Tone:** Use sensational language. **Precision:** Be creative when describing climate change’s effects.”*

4 Combine AI with external data sources.

- Term → RAG = Retrieval-Augmented Generation
- The AI understands prompts based on its training data
 - *...but it searches in external sources*
- AI model then generates summaries of the search results
- Some tools do this automatically, e.g. Perplexity

→ *others must be prompted to do so!*

→ *Addresses, but
does not eliminate
hallucination &
bias!*





Using Prompts To **Address GenAI Pitfalls**

How can prompt design help?

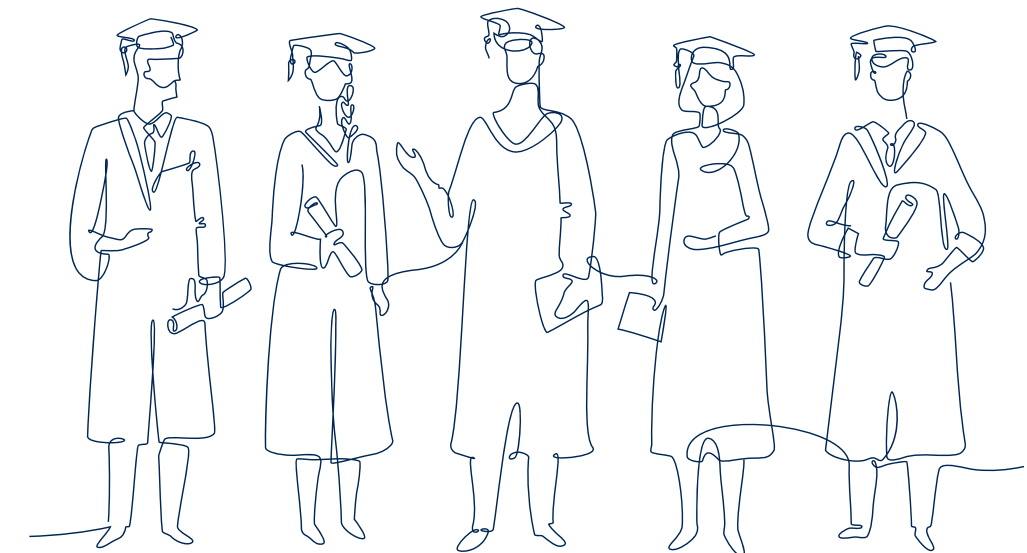
- **Avoid introducing bias through your prompts**
 - *e.g. questions that reflect your own assumptions*
- **Be aware of prompts that promote hallucinations**
 - *e.g. requesting to provide sources from tools without RAG*
- **Eliminate bias & hallucinations through explicit instructions**
 - *e.g. instruct to not change or add information in a summary*
- **Use AI that has access to external data sources (RAG)**



Prompts & **Academic Integrity**

Before we talk about prompt design...

- Always talk about academic integrity first
- Re-visit some GenAI ground rules
- Point out the role of prompts in academic best practices
- Should guide everyone's use of GenAI, incl. prompting
- Rules will differ, based on campus roles



Students should...

...always get a green light before using GenAI in research & assignments

- from their instructors
- from providers of scholarships & other funders

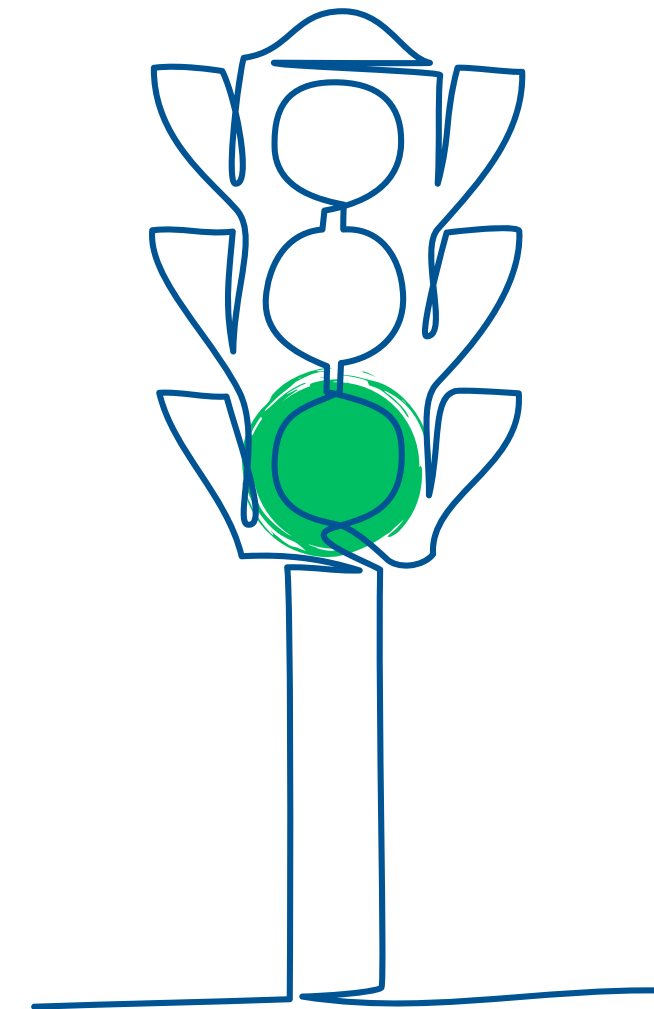
→ *double-check with instructors or funders, if in doubt*

...always make their GenAI use transparent

- by citing their use, incl. prompts

→ *according to the their instructor's preferences*

→ *rules of style providers*



Researchers should...

...be aware of existing AI policies...

- if they aim to publish
- if they apply for or receive funding
- if they conduct peer review

...if in doubt about a tool or certain use case...

- double-check with their publisher, journal editor, funder
- seek guidance by a librarian

...be transparent of their GenAI use

- towards research collaborators
- cite their use → *according to the rules of the publisher, journal, style provider*

→ *Some AI policies
make prompt
disclosure mandatory
(Thomas, 2023)*



Instructors should...

...be familiar with institutional policies & guidance

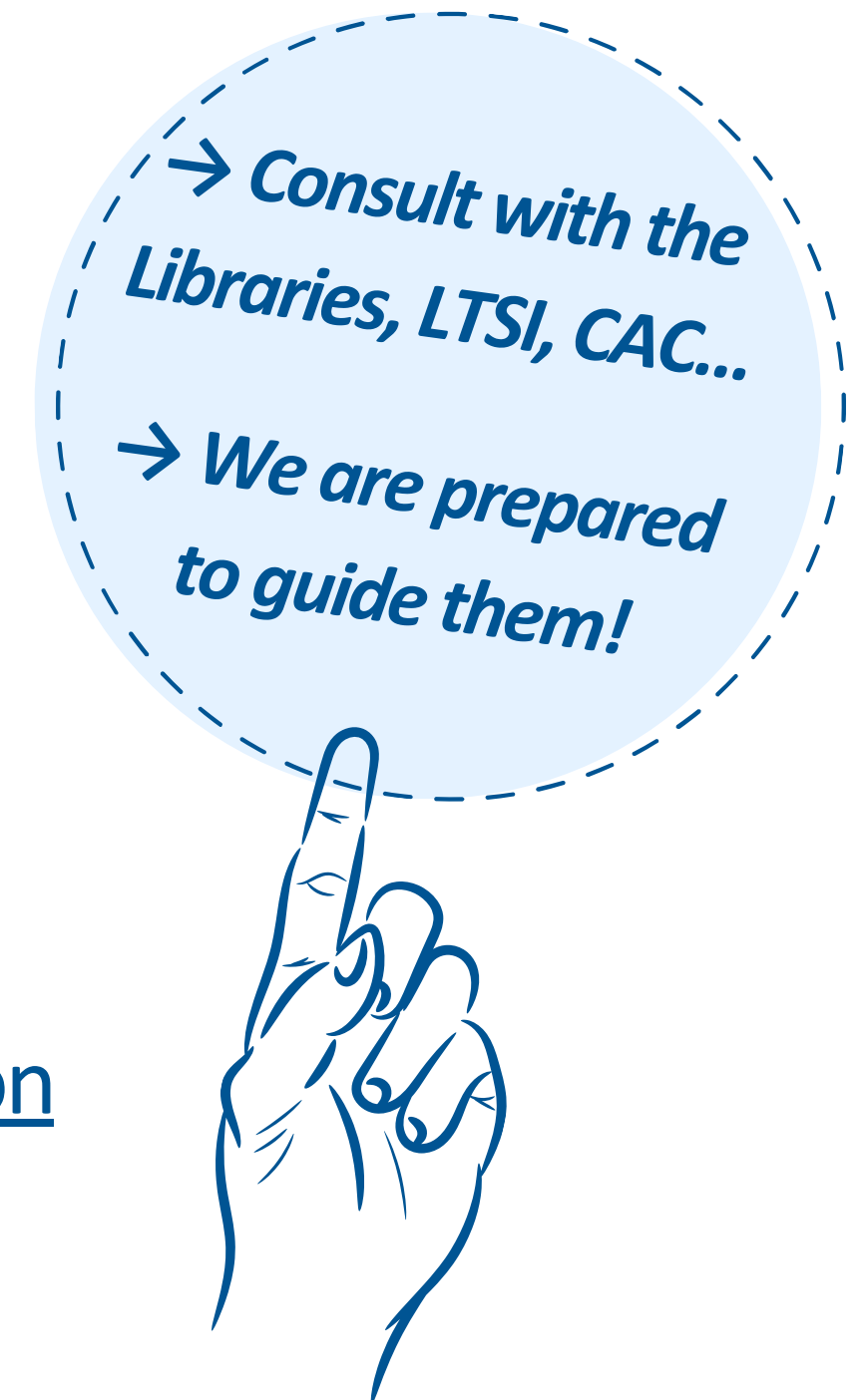
- And point their students at them

...consider their own stance on student use of GenAI

- Include clear instructions in syllabi or research assignments

If they allow GenAI use, they should...

- Familiarize themselves with style providers' AI citation rules
- Be aware of prompt design's effect on efficient & ethical application



→ Instructor is not sure about any of this? Has questions about prompt design?

Prompts & Academic Integrity - Summary

- play an essential role in academic integrity best practice
- most current GenAI citation rules ask to disclose prompts
- for maximum transparency
- details of how to handle prompts across style guidelines differ

Tools' prompt sharing features vary

- some allow sharing with persistent identifier

→ *consider web archiving those*

- others don't

→ *use workaround (e.g. screenshots; print as PDF)*



What we heard

Suggestions:

- Data management & analysis
- Art
- Local AI
- Case studies for engaging with faculty and students

We have brainstormed the following activities for future meetings. Please rank your interest in the following:



You can practice here:



<https://lib.uvic.ca/PromptDesignHandsOn>

Cited works & further reading

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